
Personal Profile

- Over 10 years' experience analysing large data-sets as a Data Scientist and as a PhD student at CERN.
- Expertise in applying data science to create insights and build data products.
- Strong communication skills; including presenting data-driven conclusions to business stakeholders.

Technical Skills

- **Computing** Python (9 years' experience) incl. data science packages; e.g. Pandas, Matplotlib, Scikit-learn. Creating data pipelines using PySpark, SQL, and automated code testing (7 years). Experienced in using Git and Cloud services (Microsoft Azure).
- **Statistics** Statistical modelling using linear regression and machine learning techniques. Hypothesis testing using A/B and synthetic-control experimentation.
- **Soft Skills** Presenting analyses; from non-technical colleagues to international scientific conferences. Leadership experience through line management and mentoring junior data scientists.

Experience & Education

Mar 2023 - **HM Treasury (HMT)**
Principal Data Scientist

- Delivery and ongoing maintenance of high-profile Financial Market monitoring dashboard.
 - Plotly Dash dashboard using Bloomberg data to summarise market moves every 15 minutes.
 - Collaborate with Market Monitoring teams to apply updates and fixes, ensuring a reliable service for critical end-of-day updates, and enabling monitoring processes to scale during recent market volatility.
 - Own Bloomberg data ingestion; responsible for managing costs, fixing bugs and improving process.
 - Dashboard relied on by 100 different users including in Ministerial and Permanent Secretary offices.
- Created new analysis service to project long-term impacts of preventative spending policies.
 - Built regression models using longitudinal survey data to project long-term impacts of policies.
 - Presented and discussed model with academics at LSE, UCL and York to ensure robust methodology.
 - Promoted service to policy teams across Government to gain buy-in and identify impactful policies.
 - Finalised projections for 2 policies, which are being used to inform the current Spending Review.
- Senior leader within team and wider analytic community.
 - Manage 2 apprentices; responsible for their development, project sourcing and oversight of delivery.
 - Proposed and run weekly retro, which is the team forum to celebrate victories and identify pain-points.
 - Gave regular training sessions to other analysts on topics ranging from using Git to Causal Inference.

Nov 2020 - **Institute for Apprenticeships and Technical Education (IfATE)**

Mar 2023 *Senior Data Scientist*

- Led analysis for T Levels, a new qualification launched in Sept. 2020
 - Built a pipeline using Python, SQL and Pandas to extract, clean and combine data from 3 sources.
 - Developed a T Level Dashboard using Python and Bokeh to enable users to create data insights.
- Developed methodology to semi-automate the matching of classification code to each apprenticeship.
 - Used Random Forest model with Word2Vec similarity scores to predict the probability of match
 - Proof-of-concept analysis shows that we can auto-assign 40% of apprenticeships with 88% accuracy.
- Line managed a junior data scientist, and mentored an aspiring data scientist.
- Proposed and ran the first ever hackathon at our organisation, encouraging collaboration and innovation.

Jan 2018 - **Booking.com**

Oct 2020 *Data Scientist - Insights*

- (Jan 2020 - Oct 2020) App Marketing product team, growing the number of app users.
 - Built data product that analysed A/B experiments for Product Manager in team.
 - Support team designing A/B experiments and understanding key business metrics for the team.
- (Jan 2018 - Dec 2019) Customer Service Operations Analytics.
 - Owned a key performance metric; created a data pipeline using PySpark to extract, clean and transform a large data-set to analyse the metric. Supported stakeholders to use and interpret the metric.
 - Determined the key factors driving a long-term deterioration of a performance metric using a linear regression model, as a result the department set-up projects to tackle and mitigate for the key factors.

Sep 2014 - **University College London and ATLAS Experiment, CERN**

Dec 2017 *PhD in Particle Physics*

- Applied statistical techniques to search for new Physics in very large data-sets from the LHC at CERN.
- Contributed to four public results and presented results at international conferences.

Sep 2010 - **Merton College, University of Oxford**

Sep 2014 *MPhys Physics*

- Achieved a 2:1 (68%), course involved mathematical and statistical problem solving for a range of situations.